

Name:

BUID:

Workshop 1: Designing Functions

1. Implement tail-recursive versions of the functions `int_of_digits` and `digits_of_int` from today's workshop. You should work on your computer but then write your final solution to `int_of_digits` on this paper.
2. Consider the following Python code for finding Pythagorean triples, i.e., (i, j, k) such that $i^2 + j^2 = k^2$.

```
def gcd(i, j):
    while i != j:
        if i > j:
            i = i - j
        else:
            j = j - i
    return i

def py_triples(n):
    out = []
    for i in range(1, n + 1):
        for j in range(i, n + 1):
            for k in range(j, n + 1):
                if i * i + j * j == k * k and gcd(i, gcd(j, k)) == 1:
                    out.append((i, j, k))
    return out
```

Determine the equivalent OCaml functions. You do not need to write your solution on this paper. Instead write either **completed**, **started** or **did not start** for your solution to indicate how far you got on the problem.

Write your solutions below and on the back on this page.

Continue your work here.